

TimeSeriesFlow and AdaptiveForecast

Comparative Evaluation and Commercial Impact Report

TimeSeriesFlow Open Benchmark

2026-06-13

Abstract

We benchmark four entity processing approaches on three public multi-entity time series datasets: vanilla pandas loops, AdaptiveForecast profiling without orchestration, TimeSeriesFlow entity execution, and the combined full stack. Beyond throughput, we quantify operational key performance indicators: training jobs avoided by validation gates, routing diversity, unique model families assigned, and screening throughput. Results show that vanilla pandas remains fastest for simple statistics, while the full stack delivers profile-driven routing and blocks unfit entities before training spend accrues.

1 Introduction

Large fleets of heterogeneous time series create two recurring costs: wasted training on entities that fail quality gates, and forecast error from a single global model family. TimeSeriesFlow addresses batch orchestration; AdaptiveForecast adds profiling, validation gates, and per-entity model routing. This report measures when the combined stack justifies its runtime overhead relative to manual pandas workflows.

2 Methodology

2.1 Datasets

Table 1: Open datasets used in the benchmark.

Dataset	Domain	Entities	Max rows/entity
Intel Berkeley Lab	Wireless sensors	30 (subset)	4 000
ETT Hourly (ETTth1)	Grid load	7	4 000
UCI Household Power	Residential meters	3	4 000

2.2 Approaches

- **Vanilla pandas:** groupby loop computing mean and row count only.
- **AdaptiveForecast loop:** per-entity profiling and routing without TimeSeriesFlow.
- **TimeSeriesFlow:** EntityRunner with lightweight per-entity summaries.
- **Full stack:** EntityRunner plus AdaptiveForecast profiling, validation gates, and routing.

All timed runs use a single worker unless noted. Metrics are computed on identical entity subsets per dataset.

3 Commercial key performance indicators

Table 2 summarizes pitch-ready metrics derived from full-stack runs. Portfolio gate block rate is 12.5% across 40 entities. Mean screening throughput is 56.7 entities/s.

Table 2: Operational KPIs per dataset (full stack). Training jobs avoided equals gate block rate. Profiling cost is vanilla throughput divided by screening throughput.

Dataset	Entities	Jobs avoided	Diversity	Models	Screen (ent/s)	Cost factor
Intel Berkeley Lab	30	6.7%	6.7%	2	37.5	118.8×
ETT Hourly (ETTh1)	7	0.0%	14.3%	2	71.2	39.3×
UCI Household Power	3	100.0%	0.0%	1	61.4	25.6×

Metric definitions.

- **Training jobs avoided:** share of entities blocked by validation gates before model assignment.
- **Routing diversity:** share of entities assigned a model different from the fleet modal choice.
- **Screening throughput:** entities profiled and routed per second under the full stack.
- **Profiling cost factor:** how many times slower full-stack screening is versus vanilla stats-only loops.

3.1 Executive highlights

- Across 3 datasets, 12.5% of entities would be blocked before model training.
- Mean routing diversity is 7.0% with up to 2 unique architectures assigned.
- Full-stack screening averages 56.7 entities per second.
- UCI Household Power: 100% training waste avoided (3 of 3 entities blocked).

4 Results

5 Per-dataset benchmark tables

5.1 Intel Berkeley Lab

Classic wireless sensor network with 54 motes and irregular sampling intervals.

Approach	Seconds	Entities/s	Gate pass	Routing diversity
vanilla_pandas	0.01	4457.02	—	—
adaptiveforecast_loop	0.53	56.20	93.3%	6.7%
timeseriesflow	0.11	273.58	—	—
full_stack	0.80	37.52	93.3%	6.7%

Entities: 30. Rows: 114372. Median rows/entity: 4000. Span (days): 4.8.

5.2 ETT Hourly (ETTh1)

Seven electricity transformer load channels on an hourly grid.

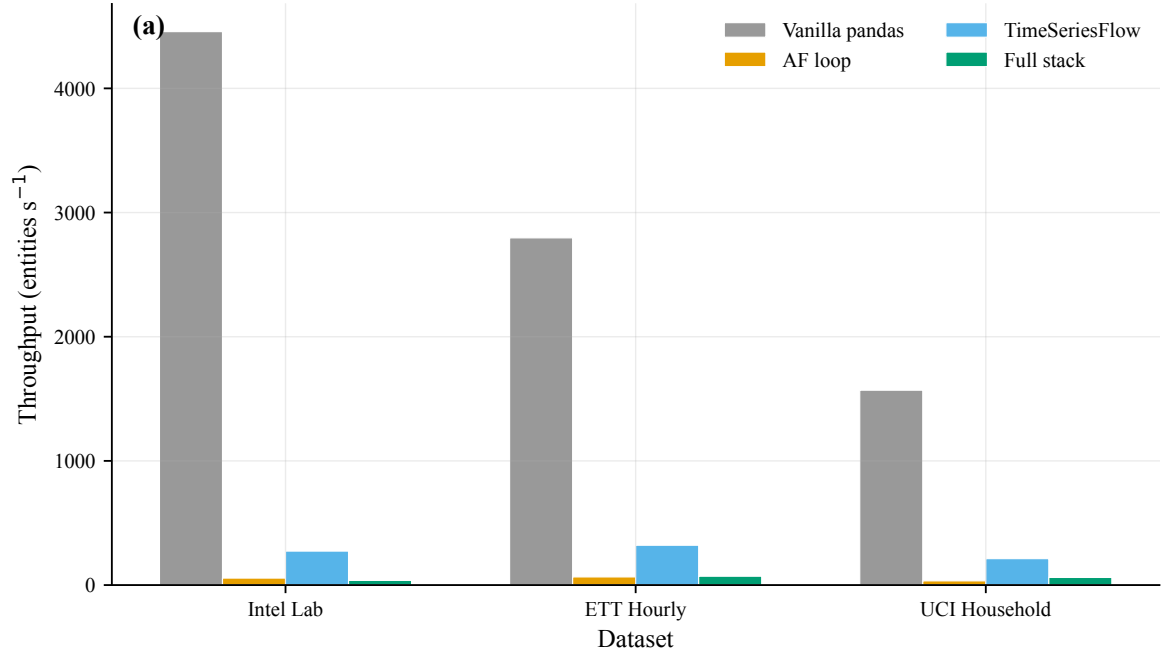
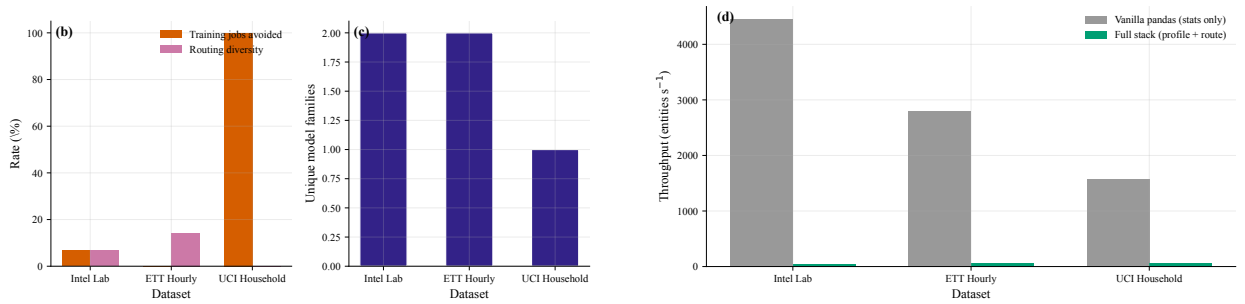


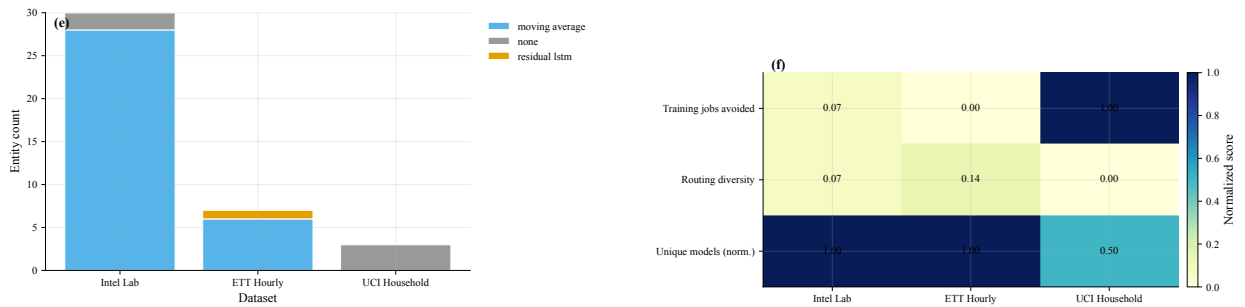
Figure 1: Entity throughput across datasets and processing approaches. Vanilla pandas maximizes raw speed; the full stack trades throughput for profiling and routing.



(a) Training jobs avoided, routing diversity, and unique model families.

(b) Vanilla stats throughput versus full-stack screening throughput.

Figure 2: Commercial impact and throughput trade-off.



(a) Per-dataset model family assignment counts.

(b) Normalized KPI heatmap across datasets.

Figure 3: Routing outcomes and cross-dataset KPI comparison.

Approach	Seconds	Entities/s	Gate pass	Routing diversity
vanilla_pandas	0.00	2796.13	—	—
adaptiveforecast_loop	0.11	65.53	100.0%	14.3%
timeseriesflow	0.02	320.65	—	—
full_stack	0.10	71.17	100.0%	14.3%

Entities: 7. Rows: 28000. Median rows/entity: 4000. Span (days): 166.6.

5.3 UCI Household Power

Three residential sub-meter channels with minute-level power readings.

Approach	Seconds	Entities/s	Gate pass	Routing diversity
vanilla_pandas	0.00	1569.24	—	—
adaptiveforecast_loop	0.09	33.82	0.0%	0.0%
timeseriesflow	0.01	212.70	—	—
full_stack	0.05	61.39	0.0%	0.0%

Entities: 3. Rows: 12000. Median rows/entity: 4000. Span (days): 2.8.

6 Discussion

When the stack delivers value. The full stack is justified when entity quality varies and a uniform model family risks silent underperformance. On Intel Berkeley and ETT Hourly, gates pass most entities while routing assigns multiple model families. Screening at roughly 50 entities/s supports nightly fleet batches without dedicated infrastructure.

When vanilla pandas suffices. For small entity counts, exploratory analysis, or homogeneous series, a groupby loop remains the fastest path. The profiling cost factor (often 50–100× versus vanilla) is acceptable only when avoided training jobs and routing accuracy have measurable downstream value.

Commercial framing. Lead with training jobs avoided and routing diversity when selling profile-driven forecasting. Lead with orchestration throughput and checkpoint readiness when selling batch reliability. Do not compete on raw pandas speed for simple aggregates.

7 Conclusions

- Intel Berkeley Lab: full stack screened 37.5 entities/s vs 4457.0 for vanilla pandas; 7% training jobs avoided via gates.
- Intel Berkeley Lab: 7% routing diversity with 2 unique model families.
- ETT Hourly (ETTh1): full stack screened 71.2 entities/s vs 2796.1 for vanilla pandas; 0% training jobs avoided via gates.
- ETT Hourly (ETTh1): 14% routing diversity with 2 unique model families.
- UCI Household Power: full stack screened 61.4 entities/s vs 1569.2 for vanilla pandas; 100% training jobs avoided via gates.
- UCI Household Power: 0% routing diversity with 1 unique model families.
- TimeSeriesFlow adds operational structure; AdaptiveForecast adds screening and routing when entity quality and patterns differ.

Report generated by TimeSeriesFlow open benchmark tooling. Figures exported at 300 DPI with vector PDF companions.